How the Past Changes What Can Happen Next

A true-layperson guide to IOTA and the Information–Constraint Principle

Douglas Clark

Plain-language companion — 20 July 2026

The answer in one paragraph. IOTA and the Information–Constraint Principle are worth pursuing as a scientific research program. The project has already shown, inside carefully defined computer models, that a useful relational role can disappear in one place and reappear through a completely different local arrangement. Other sciences show many parts of the larger idea: memories survive their initiating events, physical records alter later behavior, and organized quantum systems can produce particle-like patterns. But no current result shows that the universe is made of an IOTA lattice, that information is the unique foundation of nature, or that matter and energy have been derived from it. The idea has legs; it has not yet crossed the finish line.

The question that started the journey

How can the effect of something survive after the original thing is gone?

A foot lifts from wet ground, but the footprint remains. A bell stops moving, but a recording can reproduce its sound. A virus attacks a bacterium; much later, descendants of that bacterium can carry a molecular record of the attack. In each case, something about an earlier encounter matters later even though the original actor or arrangement is no longer present.

This can sound mysterious only because the word *information* is used for several different things. The footprint is not the foot. It is also not a ghostly message floating outside the physical world. The mud has been changed. The recording is not the bell, but a new physical arrangement carries a usable pattern. The bacterial record is not the virus, but molecules in the cell now make a different response possible.

That distinction is the starting point for IOTA:

> Information may persist *beyond its original carrier*. That does not mean it persists *without any physical carrier*.

The deeper question is whether this familiar kind of persistence is a clue to something more basic. Could reality be built, at its lowest level, from events, relationships, and rules about which events may follow? Could the things we usually treat as fundamental—particles, fields, mass, energy, space, and time—instead be durable patterns that emerge from that deeper organization?

IOTA explores that possibility. The Information–Constraint Principle, called the principle in the rest of this essay, is the proposed bridge between the basic layer and the world we know. This paper asks two questions. Does that proposal fit what science currently knows well enough to deserve serious work? And if it does, what must happen next?

The idea in ordinary language

The project's working definition is deliberately practical:

> **IOTA's premise.** Information is a change in what can happen next, caused by an interaction.

Imagine a railway yard. The same trains can reach different destinations depending on how the switches are set. The switches do not push every train to one inevitable destination. They shape the routes that are open, closed, easy, or unlikely. An earlier worker may set a switch and leave. The worker is gone, but the setting still changes the journeys available to later trains.

This is what *constraint* means here. It is not merely a restriction and it is not a new invisible force. A constraint can close a path, open a path, redirect traffic, or change the odds among paths. A bridge can make a destination reachable. A memory can make one response more likely than another. An arrangement can preserve useful options at one level while ruling out movements at another.

The phrase *what can happen next* also needs care. It never means every future anyone can imagine. It means the futures available under stated rules, conditions, actions, measuring methods, resources, and time span. A locked door is impassable to a person with bare hands, but not to a firefighter with an axe. Both statements can be true because they describe different allowed actions. Mathematics for reachable and survivable futures already exists in control and viability theory [1, 2]. Any new principle must respect and build beyond that work.

IOTA makes a larger proposal about the foundation of physics. Its candidate starting layer contains local distinctions, relationships, and rules of succession. The present computer lattice is one concrete way to test the idea. It is *not* being declared the literal grid of the cosmos. If the research program succeeds, familiar physics would have to emerge in stages:

1. interactions produce differences in later possibilities;
2. some differences persist and can pass to new physical carriers;
3. some become durable, coordinated patterns;
4. some patterns acquire particle-like behavior;
5. interacting patterns recover fields and conserved quantities; and
6. mass, energy, geometry, clocks, and ordinary large-scale physics emerge as reliable features of that organization.

This is a research ladder, not a list of completed discoveries. The present project is near its first rungs.

What the IOTA project has actually shown

The cleanest project result concerns a bridge. In the computer model, two regions were connected through a local arrangement of cells. That original bridge disappeared. Later, the regions became connected again through a new bridge that shared no cells with the first one. Removing the new bridge then destroyed the renewed connection. The new arrangement was therefore doing the work; it was not merely near the place where the work happened [3].

The numbers matter. Thirty-six candidate relocation events were replayed. Twelve had the right

kind of original bridge, and three events across two independent starting cases passed the strict replacement test. That is a real positive result, but a narrow one. It shows that this kind of role replacement *can* happen in the declared model. It does not show that it always happens, lasts forever, or occurs in nature.

Follow-up tests made the picture richer. Some relationships survived after the original cell addresses died because the project tracked a continuing line of expression rather than a fixed address. Some lasted much longer than others. Opening a blocked route did not guarantee that the route would eventually be used: timing and competing events could close the opportunity first. When an observed successor was prevented, the model often found another successor. This exposed hidden backups. A role could be necessary only after several alternatives had been removed in sequence.

These are useful lessons about possibility. An open route is not the same as a realized outcome. A thing that looks necessary may have an unseen substitute. A record tied to one address may look dead even while its role continues elsewhere. The right object may be a continuing relation, not a marked lump moving across the screen.

The project also kept its failures. One proposed way of describing a graph role did not pass its full confirmation test. A later independent repair study did not collect the minimum number of eligible cases, so its outcome was not opened and the idea remains untested. These are not embarrassing details. They are evidence that the tests can say no.

A second project asked whether two local quantum models could reproduce a limited part of a well-known equation used for electron-like particles. They did pass the surviving tests for that limited role [4]. But the construction began with quantum rules already supplied. It also supplied the clock, the rule that turns a quantum description into measurement probabilities, and a mass setting. An audit found a faulty test that had looked at an empty part of the data; those results were withdrawn. The audit also showed that the two free models were more closely related than first thought.

The honest conclusion is encouraging and restrained: local mechanisms can reproduce a bounded piece of known quantum behavior, and the audit improved the claim. This is not yet a derivation of quantum theory, a particle, or mass from base IOTA.

A third project studied the geometry of meaning in an engineered language system [5]. It showed that relations to fixed reference statements can preserve useful structure and that comparisons made from separate reference sets can be checked against one another. It also used negative controls to show what the method could not establish. This is useful evidence about relational measurement. It is not evidence that nature is made of language or that an artificial-intelligence embedding is fundamental physics.

What the wider world already teaches us

The larger scientific literature does not prove IOTA. It does show that the program is asking a coherent question and that many pieces of the proposed mechanism are physically possible.

Records can leave their source

In quantum experiments, a system can interact with its surroundings so that different pieces of the surroundings carry readable traces of the system. Under the right conditions, separate observers can learn the same limited fact by measuring different environmental pieces rather than touching the source. This area is sometimes called quantum Darwinism. Its lesson is modest but important: an

interaction can create secondary physical carriers of a record [6–10].

It does not follow that every environment makes good records, that the record exists without a carrier, or that there is always one uniquely correct way to divide the world into "system" and "environment." A correlation is not automatically an accessible memory, and a memory is not automatically a new law of nature. Modern descriptions of multi-step quantum experiments already capture how earlier interactions can influence later responses [11, 12]. The principle must add something testable beyond a complete version of that account.

Living things are made of maintained history

Biology supplies especially vivid examples. After certain bacteria survive a virus, they can insert a short piece of viral sequence into their own genetic record. Editing that record changes which virus they resist later [13]. In yeast, an earlier period of nutrient shortage can establish a chromatin state that outlasts the factor that initiated it and survives several cell divisions. Remove the machinery that maintains the mark, however, and the memory fades [14, 15].

Evolution also has a path. In a classic protein experiment, researchers built every combination of five mutations. Under the experiment's stated conditions, only a small fraction of the direct step-by-step routes kept improving at every step. Earlier changes made some later changes workable and others damaging. The result does not mean that nature had only those routes under every population, environment, and time span. It means that history changed accessibility under the declared test [16, 17].

Experiments with self-copying molecules and cell-like compartments tell a similar story. Cooperative networks can amplify one another. Boundaries can protect useful replicators from parasites. Yet self-copying, self-maintenance, heredity, and open-ended evolution are not the same achievement. Many synthetic systems still depend on researchers supplying food, enzymes, timing, or division. Biology therefore supports physically maintained, history-dependent constraint. It does not establish a new foundation of physics [18–21].

Information can change the work a machine can obtain

A controller that records where a tiny moving particle is can use that record to choose a later action and gain free energy—energy available to do useful work. This has been shown in a laboratory experiment and fits established thermodynamic accounting [22, 23]. The information is not free magic: the measuring device, memory, controller, reset procedure, heat bath, and allowed actions all matter.

A 2026 experimental preprint gives a close modern example. A visible particle in a complex fluid appeared to remember its history because hidden parts of the surrounding fluid relaxed slowly. Using delayed measurements, researchers reported extracting work associated with that hidden memory [24]. This is exciting precisely because it makes the boundary clear. The memory is outside the currently observed particle, but it remains physically carried by the environment. Once those hidden parts are included, ordinary physics may fully explain the effect.

Particle-like things can emerge from organized many-part systems

The word *particle* can suggest a tiny hard bead. Physics is subtler. Inside materials and engineered quantum systems, coordinated disturbances can behave like particles: they can travel, carry charge-like properties, collide, or obey unusual exchange rules. Physicists call these collective objects *quasiparticles*.

A stadium wave is a useful first picture, as long as we do not mistake it for a quantum theory. The wave moves around the stadium while individual people only stand and sit. The moving pattern has a life and behavior not captured by following one person. In real quantum systems, the collective behavior is far richer and is backed by exact models and controlled experiments. Local quantum models can produce fractional and particle-like excitations [25, 26]; quantum processors and atomic gases have engineered and measured striking examples [27, 28].

This defeats a blanket objection: particle-like behavior *can* emerge from organized local ingredients. It does not prove that IOTA's present patterns are particles. Every example just named already assumes quantum mechanics. A cellular quantum model can approach a limited free-field theory in one dimension, while related constructions face serious barriers in higher dimensions [29]. IOTA must build the quantum layer it currently borrows.

Four tempting clues that are not proof

Some of the most exciting subjects in science are useful to this project because they expose mistakes the project must avoid.

Zipf's law: the same shape can have many causes

Put words in order from most common to least common and their frequencies often fall in a remarkably regular way. Similar rank patterns appear in other kinds of data. This is Zipf's law. It is tempting to see such a clean curve as the signature of a deep organizing principle.

But very different mechanisms can make Zipf-like patterns: random text, growth with preferential reuse, hidden variables, least-effort models, and history-dependent shrinking of the available choices all produce related shapes under different conditions [30–34]. Even deciding whether data truly follow a power law requires careful comparison with rival distributions [35].

So a Zipf curve cannot prove IOTA. At best it can suggest a question. The useful test would hold the visible rank curve nearly the same while changing the history or the proposed carrier, then ask whether later responses differ. If two systems share the curve but not their future behavior, the pattern alone was never the mechanism.

The adjacent possible: the closest neighboring idea

Stuart Kauffman called the possibilities one step beyond what already exists the *adjacent possible*. In a chemical example, these are the new kinds that could be made by one more reaction using kinds already present [36]. This is strikingly close to the principle's question: how does today's organized physical situation change what can happen next?

The idea now has mathematical and empirical descendants. Triggered-discovery models can reproduce clustered novelty and familiar growth patterns; evolving networks make some new concepts easier to reach than others; and newer work tracks novel combinations, not just novel items [37–39]. Other models predict a quiet period followed by very rapid combinatorial growth, but that behavior depends on the model's counting rules and resources [40]. Work on scaffolding shows how an organized setting can bias which nearby possibilities become accessible, while long-run product data suggest that the existing product network predicts some—but not all—later discovery and that resource limits matter [41, 42].

The correlation with IOTA is therefore large, but it is mainly a meeting of structures, not independent

confirmation. In IOTA's bounded veto tests, preventing an observed successor often exposed another one; repeated vetoes revealed an ordered sequence of alternatives in successively changed histories. That resembles exploration of an adjacent frontier. Yet the computer model had finite states and frozen rules. The alternatives were hidden paths inside a declared system, not evidence that nature had created an unforeseeable new kind of state.

This comparison also exposes a useful accounting problem. Closing paths through a known menu is *pruning*. The adjacent possible can instead add items to the menu, or replace old possibilities with different ones. A single count can miss that difference: two menus may be the same size while sharing nothing. Future tests must therefore state what counts as a new kind, how many steps and resources are allowed, and separately record possibilities that were retained, lost, and newly opened.

Adjacent-possible research is close prior art for the broad claim that history reshapes possibility. IOTA earns something stronger only if an embodied carrier does causal work beyond that familiar claim. The sharper experiment would alter or transfer the proposed carrier, map the nearby possibilities before looking at outcomes, and test for a long-horizon crossover: fewer immediate routes but more genuinely new role-level routes later. That result would also need to survive reasonable changes in what the researchers count as a role and comparison with ordinary fixed-space and discovery models.

Entropy: one number can hide different futures

Entropy is not one universal synonym for disorder. In communication, it can measure uncertainty in a set of messages [43]. In thermodynamics, it is tied to physical states, heat, work, and the direction of large-scale change [44, 45]. Other entropies describe dynamical unpredictability, quantum resources, or coarse-grained descriptions.

Constraint and entropy are related, but neither simply replaces the other. Closing paths can lower uncertainty in one description. Maintaining a living cell can keep a narrow set of local conditions by releasing heat elsewhere. Two systems can have the same entropy value while allowing different outcomes, remembering different histories, or requiring different resources to reverse. The answer changes when we change which states count as distinct, what can be measured, what actions are allowed, and how long we wait.

For the principle, entropy is therefore a check, not a shortcut. Every claim must state the menu of states, the probabilities or physical measure, the boundary, the allowed actions, and the time span. A claim that merely renames low entropy as "more constraint" would add little. More promising is to ask whether an embodied record changes the structure of future paths even when a single entropy score is held fixed. Work connecting prediction and wasted energy shows why stored history can matter without making information a new substance [46].

Black holes: preservation is not the same as recovery

Black holes make the persistence question unavoidable. If an object falls in and the black hole later evaporates, what happens to the distinctions that once described the object? Modern theory has made major progress. Calculations in controlled gravitational models can reproduce the expected rise and fall of the radiation's entropy, often called a Page curve [47–49]. Ideas from quantum error correction also explain how information about one region can be represented in a distributed and indirect way [50].

But several questions must not be collapsed into one. Is information preserved in the complete state? Can a chosen set of measurements distinguish it? Can it be reconstructed in principle? Can an

actual observer recover a message? Can that recovery be done before the required time or machinery becomes absurd? Computational barriers can separate "present in principle" from "usable" [51].

Page curves, islands, and reconstruction maps are not identified physical carriers in the simple sense of a memory chip or a molecule. The strongest results are theoretical and often belong to carefully controlled models, not to observed evaporation of an astrophysical black hole. Black-hole information theory is therefore an exceptional stress test for the principle's language of carriers, access, boundaries, and recovery. It is not evidence that IOTA is the substrate of the universe.

The strongest skeptical answer

A critic can make a powerful and simple objection:

> If we know the complete physical state, the laws, the allowed actions, and the boundary, then ordinary physics already tells us the possible futures. Calling part of that state "information" and its effects "constraint" may only rename what was already there.

This objection is not a misunderstanding. It is the main scientific test. Control theory already calculates reachability. Predictive models already group histories that lead to the same future probabilities. Causal methods already ask what changes after an intervention. One established framework even scrambles correlations and measures whether an organism-like system remains viable [52]. The broad claim "information changes future possibilities" is not new.

For the proposed principle to become more than a useful viewpoint, carrier succession must earn at least one concrete payoff. It might reveal a smallest physical description that predicts all relevant responses. It might identify a feature that remains stable across very different physical realizations. It might prove that some transformation is impossible. It might compress many intervention results better than existing methods. Best of all, it might make a new prediction that ordinary complete-state accounts do not make.

If none of those happens, IOTA could still offer a productive vocabulary and a good way to design experiments. It would not be a distinct law of physics.

The honest verdict

The verdict is positive, but graded:

> **IOTA and the principle have real scientific legs as a research program and an operational way to connect persistent records with future possibilities. They are not established as a fundamental law or as the foundation of nature.**

That conclusion rests on more than analogy. The central question can be tested. The project has produced bounded causal demonstrations, including replacement of an original bridge by a disjoint successor. Independent sciences establish that records can be physically copied, maintained, moved, and used; that history can reshape biological and thermodynamic responses; and that local quantum organization can support particle-like roles. There is also a clear sequence of experiments that could strengthen or defeat the larger proposal.

The ceiling on the conclusion matters just as much. The positive IOTA results come from one family of computer rules and representations. The quantum work starts with quantum mechanics rather than deriving it. No external experiment has tested the full carrier-replacement criterion as an IOTA

prediction. No result derives mass, energy, space, time, or the known laws from an informational substrate.

What we know

- Earlier interactions can leave physical records that affect later responses.
- Such records can outlast the event, object, or local arrangement that first created them while remaining carried by something physical.
- Reachable futures depend on rules, state, boundary, allowed actions, resources, and time span.
- Adjacent-possible research already supplies explicit models in which reinforcement, evolving networks, scaffolding, and resource limits shape which novelties are encountered next.
- Collective quantum organization can produce genuine particle-like behavior.
- The IOTA model has produced limited, intervention-tested examples of a relational role continuing through local carrier replacement.

What we infer

- Formation, persistence, and transfer of constraint may be a useful common language across physics, biology, and complex systems.
- Carrier succession may be a better target than searching for a tiny object that literally transports a payload.
- IOTA's veto results can be read as a bounded, role-level adjacent frontier, but not yet as open-ended creation of new physical possibilities.
- A ladder from persistent relations to coherent collective objects is scientifically plausible enough to attempt.

These are synthesis judgments. They are not results proved by any one cited field.

What we do not know

- Whether information, rather than matter, energy, quantum fields, or some other structure, is the most basic layer of nature.
- Whether the current IOTA rules describe anything beyond their computer model.
- Whether IOTA can generate quantum states and measurement probabilities without putting them in by hand.
- Whether its persistent patterns can become robust particles, interact, and recover known symmetries and conservation laws.
- Whether mass, energy, geometry, and clocks can emerge from the proposed substrate.
- Whether the principle will produce a prediction unavailable to a complete ordinary physical account.

How the idea earns the next rung

The next work should be less about collecting suggestive examples and more about forcing the proposal through hard gates.

First, repeat the carrier-replacement result with different local rules, update schedules, network shapes, and descriptions. If the effect vanishes whenever the bookkeeping changes, it is probably an artifact.

Second, identify the smallest persistent pattern that survives replacement. Prepare the same visible present by different histories. Then preserve, erase, swap, scramble, or rewrite the suspected carrier while matching ordinary energy, material, timing, and control differences. Apply the same later tests. This asks whether the record itself does causal work or merely marks an uncontrolled difference. Before looking at the outcome, define the kinds being counted and map the nearby possibilities; afterward, report which were kept, lost, and newly opened. Then follow the system long enough to test the harder crossover: whether preserving or transferring the carrier closes some short-term routes while opening more role-level possibilities later, and whether erasing or scrambling it removes that effect.

Third, build the quantum layer rather than assume it. The project must explain how alternatives combine, why quantum probabilities have their familiar form, and how separated systems compose. Failure here would stop the path to our known physics even if carrier succession remained interesting on its own.

Fourth, create a true particle-like excitation. It should be possible to make and remove it, locate it within a stated accuracy, measure how it travels, combine or exchange it with another, scatter it from a controlled obstacle, identify conserved features, and show that its role survives changes of local carrier and reasonable changes of description.

Fifth, move from isolated excitations to interacting field behavior while confronting known reasons some grid-based models fail. Only after that should the project claim to derive mass-like or energy-like quantities. A supplied mass knob is not an emergent mass.

Finally, make a risky physical prediction. The strongest future experiment would state in advance where the principle and the best complete ordinary model expect different results. If careful state enlargement always removes the difference, if no stable carrier role survives representation changes, or if the quantum bridge cannot be built without hiding quantum rules in the starting machinery, the fundamental version of the program should be lowered or abandoned.

That is not a weakness. A proposal with clear ways to fail is finally becoming science.

Where this leaves us

The original question survives the review in a sharper form. We do not need to imagine information detached from the physical world. We need to understand how an interaction builds a role, how that role can move through changing physical realizations, and how its presence changes the futures available under a stated test.

Science already shows that footprints can become records, records can guide machines and organisms, and organized many-part systems can behave like new objects. IOTA has demonstrated a small but genuine beginning: under frozen rules, a relational bridge role can be lost locally and restored through

a causally necessary successor. The remaining distance is enormous. Quantum structure, particles, fields, mass, energy, geometry, and a distinguishing prediction are still ahead.

The right conclusion is neither "proved" nor "just philosophy." It is more interesting than either. The idea is coherent, testable, partly constructed, and still exposed to decisive failure. We should continue—with the evidence labels attached, the skeptical alternatives kept in the room, and every new rung earned.

References

[1] Patrick Saint-Pierre. Approximation of the viability kernel. *Applied Mathematics and Optimization*, 29:187–209, 1994. doi: 10.1007/BF01204182.

[2] Ian M. Mitchell, Alexandre M. Bayen, and Claire J. Tomlin. A time-dependent Hamilton–Jacobi formulation of reachable sets for continuous dynamic games. *IEEE Transactions on Automatic Control*, 50(7):947–957, 2005. doi: 10.1109/TAC.2005.851439.

[3] Douglas Clark. Constraint transport and history-dependent successor accessibility in IOTA: An information-first finite local physics model. Manuscript in preparation for submission; project evidence, 2026. URL https://github.com/fmertz55/iota.

[4] Douglas Clark. Frozen operational tests of a bounded dirac role across two registered local quantum updates. Manuscript in preparation; project constructive evidence, 2026. URL https://github.com/fmertz55/iota.

[5] Douglas Clark. Anchor-triangulated semantic signatures: Frame-robust neighborhood structure with tunable semantic blocking. Manuscript under review; project empirical evidence, 2026.

[6] Harold Ollivier, David Poulin, and Wojciech H. Zurek. Objective properties from subjective quantum states: Environment as a witness. *Physical Review Letters*, 93:220401, 2004. doi: 10.1103/PhysRevLett.93.220401.

[7] Fernando G. S. L. Brandão, Marco Piani, and Paweł Horodecki. Generic emergence of classical features in quantum darwinism. *Nature Communications*, 6:7908, 2015. doi: 10.1038/ncomms8908.

[8] Mario A. Ciampini et al. Experimental signature of quantum darwinism in photonic cluster states. *Physical Review A*, 98:020101, 2018. doi: 10.1103/PhysRevA.98.020101.

[9] Ming-Cheng Chen et al. Emergence of classical objectivity of quantum darwinism in a photonic quantum simulator. *Science Bulletin*, 64:580–585, 2019. doi: 10.1016/j.scib.2019.03.032.

[10] Thomas Unden et al. Revealing the emergence of classicality using nitrogen-vacancy centers. *Physical Review Letters*, 123:140402, 2019. doi: 10.1103/PhysRevLett.123.140402.

[11] Félix A. Pollock, César Rodríguez-Rosario, Thomas Frauenheim, Mauro Paternostro, and Kavan Modi. Non-markovian quantum processes: Complete framework and efficient characterization. *Physical Review A*, 97(1):012127, 2018. doi: 10.1103/PhysRevA.97.012127. URL https://arxiv.org/abs/1512.00589.

[12] Félix A. Pollock, César Rodríguez-Rosario, Thomas Frauenheim, Mauro Paternostro, and Kavan Modi. Operational markov condition for quantum processes. *Physical Review Letters*, 120 (4):040405, 2018. doi: 10.1103/PhysRevLett.120.040405. URL https://arxiv.org/abs/1801.09811.

[13] Rodolphe Barrangou et al. CRISPR provides acquired resistance against viruses in prokaryotes. *Science*, 315:1709–1712, 2007. doi: 10.1126/science.1138140.

[14] William H. Light et al. Interaction of a DNA zip code with the nuclear pore complex promotes H2A.Z incorporation and INO1 transcriptional memory. *Molecular Cell*, 40, 2010. doi: 10.1016/j.molcel.2010.09.007.

[15] Brandon Sump et al. Mitotically heritable, RNA polymerase II-independent H3K4 dimethylation stimulates INO1 transcriptional memory. *eLife*, 11:e77646, 2022. doi: 10.7554/eLife.77646.

[16] Daniel M. Weinreich, Nigel F. Delaney, Mark A. DePristo, and Daniel L. Hartl. Darwinian evolution can follow only very few mutational paths to fitter proteins. *Science*, 312:111–114, 2006. doi: 10.1126/science.1123539.

[17] Michael J. Harms and Joseph W. Thornton. Historical contingency and its biophysical basis in glucocorticoid receptor evolution. *Nature*, 512:203–207, 2014. doi: 10.1038/nature13410.

[18] Tracey A. Lincoln and Gerald F. Joyce. Self-sustained replication of an RNA enzyme. *Science*, 323:1229–1232, 2009. doi: 10.1126/science.1167856.

[19] Vera Vasas, E"ors Szathmáry, and Mauro Santos. Lack of evolvability in self-sustaining autocatalytic networks constraints metabolism-first scenarios for the origin of life. *Proceedings of the National Academy of Sciences*, 107, 2010. doi: 10.1073/pnas.0912628107.

[20] Nilesh Vaidya et al. Spontaneous network formation among cooperative RNA replicators. *Nature*, 491:72–77, 2012. doi: 10.1038/nature11549.

[21] Norikazu Ichihashi et al. Darwinian evolution in a translation-coupled RNA replication system within a cell-like compartment. *Nature Communications*, 4:2494, 2013. doi: 10.1038/ncomms3494.

[22] Takahiro Sagawa and Masahito Ueda. Generalized Jarzynski equality under nonequilibrium feedback control. *Physical Review Letters*, 104:090602, 2010. doi: 10.1103/PhysRevLett.104.090602.

[23] Shoichi Toyabe, Takahiro Sagawa, Masahito Ueda, Eiro Muneyuki, and Masaki Sano. Experimental demonstration of information-to-energy conversion and validation of the generalized Jarzynski equality. *Nature Physics*, 6:988–992, 2010. doi: 10.1038/nphys1821.

[24] L. Muruga, F. Ginot, Sarah A. M. Loos, and Clemens Bechinger. Extracting work from hidden degrees of freedom, 2026. URL https://arxiv.org/abs/2603.06160.

[25] Michael A. Levin and Xiao-Gang Wen. String-net condensation: A physical mechanism for topological phases. *Physical Review B*, 71:045110, 2005. doi: 10.1103/PhysRevB.71.045110.

[26] Alexei Kitaev. Anyons in an exactly solved model and beyond. *Annals of Physics*, 321(1):2–111, 2006. doi: 10.1016/j.aop.2005.10.005.

[27] Mohsin Iqbal, Nathanan Tantivasadakarn, Ruben Verresen, et al. Non-abelian topological order and anyons on a trapped-ion processor. *Nature*, 626:505–511, 2024. doi: 10.1038/s41586-023-06934-4.

[28] Sudipta Dhar, Botao Wang, Milena Horvath, Amit Vashisht, Yi Zeng, Mikhail B. Zvonarev, Nathan Goldman, Yanliang Guo, Manuele Landini, and Hanns-Christoph Nägerl. Observing anyonization of bosons in a quantum gas. *Nature*, 642:53–57, 2025. doi: 10.1038/s41586-025-09016-9.

[29] Leonard Mlodinow and Todd A. Brun. Quantum field theory from a quantum cellular automaton in one spatial dimension and a no-go theorem in higher dimensions. *Physical Review A*, 102: 042211, 2020. doi: 10.1103/PhysRevA.102.042211.

[30] Herbert A. Simon. On a class of skew distribution functions. *Biometrika*, 42(3/4):425–440, 1955. doi: 10.1093/biomet/42.3-4.425.

[31] Wentian Li. Random texts exhibit zipf's-law-like word frequency distribution. *IEEE Transactions on Information Theory*, 38(6):1842–1845, 1992. doi: 10.1109/18.165464.

[32] Ramon Ferrer i Cancho and Ricard V. Solé. Least effort and the origins of scaling in human language. *Proceedings of the National Academy of Sciences*, 100(3):788–791, 2003. doi: 10.1073/pnas.0335980100.

[33] David J. Schwab, Ilya Nemenman, and Pankaj Mehta. Zipf's law and criticality in multivariate data without fine-tuning. *Physical Review Letters*, 113:068102, 2014. doi: 10.1103/PhysRevLett.113.068102. URL https://arxiv.org/abs/1310.0448.

[34] Bernat Corominas-Murtra, Rudolf Hanel, and Stefan Thurner. Understanding scaling through history-dependent processes with collapsing sample space. *Proceedings of the National Academy of Sciences*, 112(17):5348–5353, 2015. doi: 10.1073/pnas.1420946112. URL https://arxiv.org/abs/1407.2775.

[35] Aaron Clauset, Cosma Rohilla Shalizi, and M. E. J. Newman. Power-law distributions in empirical data. *SIAM Review*, 51(4):661–703, 2009. doi: 10.1137/070710111. URL https://arxiv.org/abs/0706.1062.

[36] Stuart A. Kauffman. *Investigations*. Oxford University Press, New York, 2000. doi: 10.1093/oso/9780195121049.001.0001.

[37] Francesca Tria, Vittorio Loreto, Vito D. P. Servedio, and Steven H. Strogatz. The dynamics of correlated novelties. *Scientific Reports*, 4:5890, 2014. doi: 10.1038/srep05890.

[38] Iacopo Iacopini, Staša Milojević, and Vito Latora. Network dynamics of innovation processes. *Physical Review Letters*, 120(4):048301, 2018. doi: 10.1103/PhysRevLett.120.048301.

[39] Gabriele Di Bona, Alessandro Bellina, Giordano De Marzo, Angelo Petralia, Iacopo Iacopini, and Vito Latora. The dynamics of higher-order novelties. *Nature Communications*, 16:393, 2025. doi: 10.1038/s41467-024-55115-y.

[40] Marina Cortês, Stuart A. Kauffman, Andrew R. Liddle, and Lee Smolin. The TAP equation: Evaluating combinatorial innovation. *European Economic Review*, 179:105144, 2025. doi: 10.1016/j.euroecorev.2025.105144.

[41] James Griesemer. Modeling scaffolded development into the adjacent possible environment. *Philosophy of Science*, 91(5):1404–1414, 2024. doi: 10.1017/psa.2023.144.

[42] Josef Taalbi. Long-run patterns in the discovery of the adjacent possible. *Industrial and Corporate Change*, 35(1):123–149, 2026. doi: 10.1093/icc/dtaf028.

[43] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27: 379–423, 623–656, 1948. doi: 10.1002/j.1538-7305.1948.tb01338.x. Part II DOI: 10.1002/j.1538-7305.1948.tb00917.x.

[44] Ludwig Boltzmann. On the relationship between the second fundamental theorem of the mechanical theory of heat and probability calculations regarding the conditions for thermal equilibrium. *Sitzungsberichte der Kaiserlichen Akademie der Wissenschaften*, 76:373–435, 1877. English translation by Kim Sharp and Franz Matschinsky, Entropy 17, 1971–2009 (2015), DOI 10.3390/e17041971.

[45] Elliott H. Lieb and Jakob Yngvason. The physics and mathematics of the second law of thermodynamics. *Physics Reports*, 310(1):1–96, 1999. doi: 10.1016/S0370-1573(98)00082-9.

[46] Susanne Still, David A. Sivak, Anthony J. Bell, and Gavin E. Crooks. Thermodynamics of prediction. *Physical Review Letters*, 109:120604, 2012. doi: 10.1103/PhysRevLett.109.120604.

[47] Don N. Page. Information in black hole radiation. *Physical Review Letters*, 71:3743–3746, 1993. doi: 10.1103/PhysRevLett.71.3743.

[48] Ahmed Almheiri, Netta Engelhardt, Donald Marolf, and Henry Maxfield. The entropy of bulk quantum fields and the entanglement wedge of an evaporating black hole. *Journal of High Energy Physics*, 2019(12):063, 2019. doi: 10.1007/JHEP12(2019)063.

[49] Geoff Penington, Stephen H. Shenker, Douglas Stanford, and Zhenbin Yang. Replica wormholes and the black hole interior. *Journal of High Energy Physics*, 2022(03):205, 2022. doi: 10.1007/JHEP03(2022)205.

[50] Ahmed Almheiri, Xi Dong, and Daniel Harlow. Bulk locality and quantum error correction in AdS/CFT. *Journal of High Energy Physics*, 2015(04):163, 2015. doi: 10.1007/JHEP04(2015)163.

[51] Daniel Harlow and Patrick Hayden. Quantum computation vs. firewalls. *Journal of High Energy Physics*, 2013(06):085, 2013. doi: 10.1007/JHEP06(2013)085.

[52] Artemy Kolchinsky and David H. Wolpert. Semantic information, autonomous agency and non-equilibrium statistical physics. *Interface Focus*, 8:20180041, 2018. doi: 10.1098/rsfs.2018.0041.